AEGENT

Smart Contract

Security Audit Report

30 July 2026 · Authored by AEGENT Security Team

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Executive Summary

ALL TRACKED CODE FINDINGS VERIFIED AS RESOLVED

Assessment result

No Critical or High issue was identified. All tracked code findings have verified remediation in the current source snapshot. Mainnet activation remains subject to deployment-address, role, reserve, RPC and funded-wallet acceptance gates.

Scope

Five core contracts and six production interfaces, 3,287 production lines of Solidity, plus deployment and keeper controls.

Verification

34 Solidity files compiled; 74 chain-56 tests passed with one intentional wrong-chain case skipped by design; the chain-97 guard passed independently.

Audit Methodology

Manual analysis

Line-by-line control-flow, access-control, accounting, oracle, custody, replay, reentrancy, precision and denial-of-service analysis.

Automated analysis

Hardhat compilation and tests, solidity-coverage, Slither 0.11.5 and production dependency advisory checks.

Adversarial scenarios

Stale quotes, stale or future oracle rounds, reserve shortfall, split redemption, nonce replay, wrong chain, reentrancy and fee-on-transfer assets.

Project Overview

Project

AEGENT is a BNB Chain smart-contract system for phased AGNT distribution, wallet cost-basis redemption, locked sale proceeds and term-based staking.

Security objective

Protect user value under stale dependencies, adversarial transaction ordering, reserve stress, replay attempts and privileged operations.

Project Introduction

Market lifecycle

A 90-day schedule applies fixed 10/7/3 AGNT per USDT phase rates and explicit market modes.

Core workflows

Users purchase with BNB, USDT or USDC; redeem against actual cost basis; and create funded staking positions that settle in AGNT.

Deployment boundary

Only byte-identical chain-56 contracts bound to the printed scope root are covered.

Vulnerability Information

0 CRITICAL - 0 HIGH - 9 RESOLVED

Critical / High

0 Critical and 0 High findings.

Medium

4 Medium findings tracked; all code remediation verified.

Low / Informational

1 Low and 4 Informational findings tracked; all remediation verified.

Residual activation risk

Production deployment evidence is intentionally outside the code snapshot and remains a release gate rather than an open code vulnerability.

Code Overview

Language and ecosystem

Solidity 0.8.24, Hardhat 2.29.0, OpenZeppelin Contracts 5.1.0, BNB Chain mainnet chain ID 56.

Core contracts

AegentMarketRegistry, AegentSwapV2, AegentRedemptionV2, AegentSaleProceedsVault and AegentStakingV1.

Source identity

Scope root SHA-256: 5c51c8e5df9126e3b0a5ce15e98869f6130e43bb13e4caaebf30853286763f0c. Evidence run: 20260730T020710Z.

Contracts Description

Registry

AegentMarketRegistry is the authoritative schedule, mode, endpoint and configuration-version source.

Distribution and refund

AegentSwapV2 validates payments and reserves; AegentRedemptionV2 refunds actual wallet cost basis.

Custody

AegentSaleProceedsVault locks proceeds; AegentStakingV1 escrows principal value until AGNT claim.

Visibility Description

External entry points

Purchase, redemption and staking calls validate chain, quote, nonce, market mode, asset, value and deadline.

Privileged entry points

Configuration, feed, role, pause and withdrawal operations are owner restricted and covered by explicit invariants.

Vulnerability Summary

9 OF 9 TRACKED FINDINGS RESOLVED

Critical / High

0 Critical and 0 High findings.

Medium

4 Medium findings tracked; all code remediation verified.

Low / Informational

1 Low and 4 Informational findings tracked; all remediation verified.

Residual activation risk

Production deployment evidence is intentionally outside the code snapshot and remains a release gate rather than an open code vulnerability.

Audit Result

ASSESSMENT COMPLETE

Result

No Critical or High issue was identified. Four Medium, one Low and four Informational findings are resolved in the current source snapshot.

Verification

Compilation, tests, wrong-chain validation, coverage, Slither triage and production dependency checks completed.

MED-01 - Cumulative redemption threshold

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with cumulative redemption threshold could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Wallet-level cumulative UTC-day accounting now evaluates the next total before the hard limit and manual threshold.

Validation

Targeted regression tests pass for split transactions, threshold equality and hard-limit enforcement.

MED-02 - Redemption reserve coverage

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with redemption reserve coverage could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

BNB, USDT and USDC purchase execution now fails closed unless the refund reserve covers projected cost-basis liability.

Validation

Fresh tests cover reserve readability, liability growth and insufficient-reserve rejection.

MED-03 - Persistent oracle circuit breaker

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with persistent oracle circuit breaker could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Per-feed maximum age, absolute bounds, deviation limits and permissionless persistent purchase pause are implemented.

Validation

Breaker reason classification, healthy-feed rejection and post-launch redemption continuity tests pass.

MED-04 - Staking oracle and operational controls

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with staking oracle and operational controls could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Staking validates round completeness, timestamps, decimals, asset/feed collisions, quote lifetime, slippage and inventory.

Validation

Constructor, quote, custody, reentrancy and keeper safety suites pass; production activation remains gated by deployment controls.

LOW-01 - Redemption rate-limit semantics

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with redemption rate-limit semantics could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

The enforced cumulative daily wallet limit and manual threshold prevent transaction splitting and bound wallet throughput.

Validation

Boundary tests pass; the chosen UTC-day policy is documented as the intended control.

INFO-01 - Event evidence completeness

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with event evidence completeness could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Canonical transaction, event, nonce, position and block-hash evidence is persisted for settlement workflows.

Validation

Duplicate and drift conditions fail closed in the service verification layer.

INFO-02 - NatSpec and interface clarity

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with natspec and interface clarity could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

External interfaces and security-sensitive configuration fields are documented and version-bound.

Validation

Compilation succeeds across 34 Solidity files with the current interfaces.

INFO-03 - Deployment reproducibility

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with deployment reproducibility could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

The deployment script validates the exact 90-day schedule, chain 56, feed constraints, predicted addresses and constructor events.

Validation

Eight deployment-safety regression cases pass.

INFO-04 - Operational runbook controls

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with operational runbook controls could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Keeper configuration, irreversible broadcast confirmation and public-artifact secret stripping are encoded as fail-closed checks.

Validation

Keeper and deployment tests reject missing, malformed, testnet and secret-bearing inputs.

Finding Closure Matrix

9 OF 9 TRACKED FINDINGS RESOLVED

Medium

MED-01 through MED-04: RESOLVED.

Low

LOW-01: RESOLVED.

Informational

INFO-01 through INFO-04: RESOLVED.

Compilation Results

COMPILATION - PASS

Compiler

Solidity 0.8.24, optimizer 500 runs, EVM target paris.

Result

34 Solidity files compiled successfully with zero compiler errors.

Test Results

TESTS - PASS

Chain 56

74 passing tests; one wrong-chain constructor case skipped by design.

Chain 97

Independent wrong-chain constructor guard passed 1/1.

Coverage and Static Analysis

Coverage

Production: 96.02% statements, 59.30% branches, 97.09% functions and 86.25% lines.

Static analysis

Slither 0.11.5 completed; 71 raw detector records were manually triaged.

Dependencies

Production npm advisory count: 0.

Conclusion

ALL TRACKED CODE FINDINGS RESOLVED

Code status

All nine tracked findings are resolved in the current source snapshot.

Release condition

Mainnet activation remains subject to verified deployment addresses, roles, reserves, RPC and funded-wallet acceptance.

Statement

Technical scope

This document records a point-in-time assessment of the identified AEGENT source.

Limitations

Testing cannot guarantee the absence of every defect or the security of future changes and external systems.

Advice

Nothing in this document is financial, legal, tax or investment advice.